VPRO SKILLS

FAQ / INTERVIEW PREPARATION

LLM — Scenario Based (Experienced Level) Interview Questions & Answers

STUDENT LEARNING RESOURCE

Instagram: vpro.skill. | YouTube: @vpro.skills
Contact: 9010001847

Contents

LLM Fundamentals — Real-Time Scenarios 4

RAG and Vector Database Scenarios 5

Prompt Engineering and LLM Behavior 7

Model Performance and Production Scenarios 8

Fine-Tuning, Transformers and Model Selection 9

Advanced Scenarios — Experienced Candidates 10

Quick Revision — Scenario to Solution 11

Interview tip 12

Quick Revision 13

How to Use This Guide

These questions are designed for interviews on Large Language Models (LLMs), Generative AI, RAG, embeddings, vector databases, prompt engineering, hallucinations, and production applications.

Each question includes a real-time scenario, a simple explanation, and an interview-ready answer.

This handbook brings together 27 curated questions and answers on LLM — Scenario Based (Experienced Level), organized into sections for focused revision. Each question is numbered for easy reference, and answers are broken down into definitions, examples, code, and comparison tables wherever the source material provides them.

- Use the Contents page to jump straight to a section.
- Code blocks are shown in a monospaced, shaded panel — read them alongside the explanation above.
- Comparison tables summarize key differences at a glance; use them for quick revision before an interview.
- The Quick Revision page at the end lists every question so you can self-test your recall.

01

LLM FUNDAMENTALS — REAL-TIME SCENARIOS

Q01. Your chatbot gives different answers when you ask the same question twice. Why?

SIMPLE EXPLANATION

LLMs generate answers by predicting the next token. Some configurations allow randomness.

INTERVIEW ANSWER

The output may change because of temperature, sampling strategies, context, or model nondeterminism. I would check the temperature, top-p, prompt, and model configuration. Lower temperature generally produces more consistent answers.

Q02. You give an LLM a 500-page document, but it misses information from the middle. Why?

SIMPLE EXPLANATION

The model may not use all information in a long context equally well.

INTERVIEW ANSWER

This can happen because of context-window limits, information dilution, or the “lost in the middle” problem. I would use document chunking, retrieval, relevant context selection, and focused prompts instead of passing the entire document blindly.

Q03. Your LLM generates a fake API name or invents a fact. How do you solve this?

SIMPLE EXPLANATION

This is called hallucination.

INTERVIEW ANSWER

I would ground the answer in trusted data using RAG, instruct the model to acknowledge missing information, validate important outputs, and evaluate factual accuracy. RAG reduces hallucinations but does not eliminate them completely.

Q04. A company wants its LLM to answer questions about internal company policies. Should you train a new LLM?

SIMPLE EXPLANATION

Usually, no. The company needs its documents to be searchable and available to the model.

INTERVIEW ANSWER

I would start with RAG. I would retrieve relevant policy documents and provide them to the LLM as context. Fine-tuning is more suitable when we need to change behavior, style, or task-specific patterns rather than frequently changing factual knowledge.

Q05. Your application needs answers in a fixed JSON format, but the LLM returns normal paragraphs. What would you do?

SIMPLE EXPLANATION

The model is not following the required output structure.

INTERVIEW ANSWER

I would use structured output or JSON-schema-constrained generation if supported, provide a clear schema, validate the result using a JSON parser or Pydantic, and retry or handle validation failures safely.

02
RAG AND VECTOR DATABASE SCENARIOS

Q06. Your RAG application retrieves irrelevant documents even though the correct document exists. What will you check?

SIMPLE EXPLANATION

The search system is finding the wrong information.

INTERVIEW ANSWER

I would check chunk size and overlap, embedding model suitability, query preprocessing, metadata filters, similarity thresholds, and top-k settings. I would also consider hybrid search and reranking to improve retrieval quality.

Q07. You changed your embedding model, and Pinecone returns a dimension mismatch error. Why?

SIMPLE EXPLANATION

The new model produces vectors of a different length.

INTERVIEW ANSWER

The embedding dimension must match the vector index dimension. I would create a compatible index, regenerate embeddings for all stored documents using the new model, and reindex them. I would not mix incompatible embedding spaces in the same index.

Q08. Your company has 10 million documents. Would you store all documents directly in an LLM prompt?

SIMPLE EXPLANATION

No. The prompt has context limits, and sending everything is expensive.

INTERVIEW ANSWER

I would store document embeddings in a scalable vector database, retrieve the most relevant chunks for each query, optionally rerank them, and send only the selected context to the LLM.

Q09. A user asks, “What is the refund policy?” Your vector search returns a document about the cancellation policy. What is the problem?

SIMPLE EXPLANATION

Related words do not always mean the same thing.

INTERVIEW ANSWER

Semantic similarity alone may retrieve related but incorrect content. I would improve chunking and metadata, use hybrid retrieval, add reranking, and test retrieval against a set of questions with known relevant documents.

Q10. You have a private HR chatbot. An employee asks for another employee’s salary details. How do you prevent access?

SIMPLE EXPLANATION

Retrieving a document does not mean every user is allowed to see it.

INTERVIEW ANSWER

I would enforce authentication and authorization before retrieval, filter documents by user permissions or tenant, and validate access on the server. I would never rely only on the LLM prompt to protect confidential information.

03
PROMPT ENGINEERING AND LLM BEHAVIOR

Q11. Your prompt is very long, but the answer quality is poor. What would you do?

INTERVIEW ANSWER

I would remove unnecessary instructions, clarify the task, specify the expected output format, provide a few relevant examples, and test different prompt versions against a fixed evaluation dataset.

Q12. The LLM follows a user's instruction to ignore system rules. How would you handle this?

INTERVIEW ANSWER

This may be a prompt-injection attack. I would separate trusted instructions from untrusted content, treat retrieved documents and user inputs as untrusted, limit tool permissions, validate tool arguments, and apply security checks before executing actions.

Q13. Your model provides a correct answer but uses too many words. How can you control it?

INTERVIEW ANSWER

I would define a concise response format, specify a word or token budget, provide examples of the expected style, and evaluate whether the model follows the length constraint. I would also limit the generation token budget where appropriate.

Q14. A user asks a question that is not covered by your company documents. What should the chatbot do?

INTERVIEW ANSWER

The system should acknowledge that the available sources do not contain sufficient information instead of inventing an answer. I would implement retrieval-confidence checks, source grounding, and a fallback response.

04
MODEL PERFORMANCE AND PRODUCTION SCENARIOS

Q15. Your LLM application takes 15 seconds to answer a simple question. How do you reduce latency?

INTERVIEW ANSWER

I would measure latency across retrieval, reranking, network calls, and generation. Then I would optimize prompts, reduce unnecessary context, use caching where appropriate, stream responses, and select a faster model if quality requirements permit.

Q16. Your LLM API bill increases significantly after launch. What would you investigate?

INTERVIEW ANSWER

I would track input and output tokens, requests per user, repeated queries, and model costs. I would optimize context length, cache repeatable responses, route simpler tasks to cheaper models, and apply rate limits.

Q17. Your LLM application works locally but fails in production. What is your debugging approach?

INTERVIEW ANSWER

I would inspect production logs, environment variables, API credentials, network access, timeouts, rate limits, model availability, dependency versions, and input differences. I would add monitoring, tracing, and safe error handling.

Q18. The LLM provider is unavailable. How can you keep your application usable?

INTERVIEW ANSWER

I would use timeouts, bounded retries with exponential backoff, graceful error messages, and a fallback model or provider when feasible. I would also monitor failures and avoid retry storms.

05
FINE-TUNING, TRANSFORMERS AND MODEL SELECTION

Q19. Your organization needs an LLM to classify customer complaints into five categories. Would you use a large model for every request?

INTERVIEW ANSWER

Not necessarily. I would establish a baseline using a small language model or conventional classifier, then compare accuracy, latency, and cost. I would use a larger model only if the simpler approach does not meet the requirements.

Q20. You want a model to consistently follow a particular response style and domain-specific task format. Would you use RAG or fine-tuning?

INTERVIEW ANSWER

I would choose based on the problem. RAG supplies changing or reference-specific knowledge. Fine-tuning teaches task patterns, terminology usage, or response style through examples. They can also be combined when both needs exist.

Q21. Your fine-tuned model performs well on training examples but poorly on new examples. What could be wrong?

INTERVIEW ANSWER

This may indicate overfitting, poor data diversity, or a mismatch between training and real-world inputs. I would evaluate on held-out data, improve dataset quality and coverage, tune hyperparameters, and compare against the base model.

Q22. A company wants to run an LLM on its own servers without sending data to an external API. What would you recommend?

INTERVIEW ANSWER

I would evaluate open-weight models that can be deployed locally, considering hardware requirements, model license, quantization, throughput, and accuracy. I would implement access control, monitoring, and secure storage around the deployment.

06
ADVANCED SCENARIOS — EXPERIENCED CANDIDATES

Q23. Your RAG chatbot gives a fluent answer with citations, but the cited documents do not support the answer. How do you fix it?

INTERVIEW ANSWER

I would verify that citations are tied to the actual retrieved passages, check whether each claim is supported by its source, and evaluate faithfulness separately from answer relevance. I would add claim-level citation validation and abstention when evidence is insufficient.

Q24. An agent repeatedly calls the same tool and never completes its task. What would you do?

INTERVIEW ANSWER

I would implement a maximum iteration count, execution timeout, tool-result validation, duplicate-call detection, and explicit stop conditions. I would also track each agent step to identify loops and incorrect tool decisions.

Q25. Your AI agent can read emails and send payments. How would you make it safe?

INTERVIEW ANSWER

I would follow least-privilege access, separate read and write permissions, validate every action, require human approval for high-impact transactions, maintain audit logs, and use idempotency controls to prevent duplicate actions. The LLM should not have unrestricted authority.

Q26. Your LLM passes all tests, but users complain about incorrect answers after a model upgrade. What will you do?

INTERVIEW ANSWER

I would compare the new model against the previous version using a fixed regression dataset, check prompt compatibility and retrieval behavior, measure task-specific quality and latency, and roll back if the release fails acceptance criteria.

Q27. How would you evaluate a production RAG application?

INTERVIEW ANSWER

I would measure retrieval quality using metrics such as Precision@k and Recall@k, answer quality using correctness and relevance, and grounding using faithfulness. I would also measure latency, cost, failure rate, security, and user feedback.

07
QUICK REVISION — SCENARIO TO SOLUTION

Scenario	Best starting point
LLM invents facts	Grounding, RAG, validation
Wrong documents retrieved	Improve chunking, retrieval, reranking
New embedding dimension mismatch	Compatible index and re-embedding
Answers vary	Check temperature and sampling
High API costs	Token optimization, caching, model routing
Slow responses	Trace latency and optimize bottlenecks
Need current company knowledge	RAG
Need consistent task behavior	Prompting, then evaluate fine-tuning
Sensitive documents exposed	Authorization before retrieval
Prompt injection	Trust boundaries and tool controls
Agent loops endlessly	Iteration limits and stop conditions
Model upgrade reduces quality	Regression testing and rollback

08
INTERVIEW TIP

For experienced-level interviews, answer every scenario in this order:

1. Identify the problem — Explain what may be happening.
2. Investigate — Explain what logs, metrics, or data you would inspect.
3. Implement the solution — Describe the technical changes.
4. Validate — Explain how you would test that the fix works.

This approach demonstrates practical engineering experience instead of only theoretical knowledge.

Quick Revision

Scan through these questions to self-test your recall before an interview.

Q01. Your chatbot gives different answers when you ask the same question twice. Why?
Q02. You give an LLM a 500-page document, but it misses information from the middle. Why?
Q03. Your LLM generates a fake API name or invents a fact. How do you solve this?
Q04. A company wants its LLM to answer questions about internal company policies. Should you train a new LLM?
Q05. Your application needs answers in a fixed JSON format, but the LLM returns normal paragraphs. What would you do?
Q06. Your RAG application retrieves irrelevant documents even though the correct document exists. What will you check?
Q07. You changed your embedding model, and Pinecone returns a dimension mismatch error. Why?
Q08. Your company has 10 million documents. Would you store all documents directly in an LLM prompt?
Q09. A user asks, “What is the refund policy?” Your vector search returns a document about the cancellation policy. What is the problem?
Q10. You have a private HR chatbot. An employee asks for another employee’s salary details. How do you prevent access?
Q11. Your prompt is very long, but the answer quality is poor. What would you do?
Q12. The LLM follows a user's instruction to ignore system rules. How would you handle this?
Q13. Your model provides a correct answer but uses too many words. How can you control it?
Q14. A user asks a question that is not covered by your company documents. What should the chatbot do?
Q15. Your LLM application takes 15 seconds to answer a simple question. How do you reduce latency?
Q16. Your LLM API bill increases significantly after launch. What would you investigate?
Q17. Your LLM application works locally but fails in production. What is your debugging approach?
Q18. The LLM provider is unavailable. How can you keep your application usable?
Q19. Your organization needs an LLM to classify customer complaints into five categories. Would you use a large model for every request?
Q20. You want a model to consistently follow a particular response style and domain-specific task format. Would you use RAG or fine-tuning?
Q21. Your fine-tuned model performs well on training examples but poorly on new examples. What could be wrong?
Q22. A company wants to run an LLM on its own servers without sending data to an external API. What would you recommend?
Q23. Your RAG chatbot gives a fluent answer with citations, but the cited documents do not support the answer. How do you fix it?
Q24. An agent repeatedly calls the same tool and never completes its task. What would you do?
Q25. Your AI agent can read emails and send payments. How would you make it safe?
Q26. Your LLM passes all tests, but users complain about incorrect answers after a model upgrade. What will you do?
Q27. How would you evaluate a production RAG application?